

PROCEEDINGS OF SPIE

[SPIDigitalLibrary.org/conference-proceedings-of-spie](https://spiedigitallibrary.org/conference-proceedings-of-spie)

Verification of customized conjunctival blood flow velocity measurement system with artificial motion

Hang Chan Jo, Hyun-Ji Lee, Sang-Won Lee, Dae Yu Kim

Hang Chan Jo, Hyun-Ji Lee, Sang-Won Lee, Dae Yu Kim, "Verification of customized conjunctival blood flow velocity measurement system with artificial motion," Proc. SPIE 12387, Optical Diagnostics and Sensing XXIII: Toward Point-of-Care Diagnostics, 1238709 (16 March 2023); doi: 10.1117/12.2648652

SPIE.

Event: SPIE BiOS, 2023, San Francisco, California, United States

Verification of Customized Conjunctival Blood Flow Velocity Measurement System with artificial motion

Hang Chan Jo ^{a, b, c}, Hyun-Ji Lee ^{d, e}, Sang-Won Lee ^{d, e}, and Dae Yu Kim ^{a, b, c}

^a Dept. of Electrical and Computer Engineering, Inha University; ^b Center for Sensor Systems, Inha University; ^c Inha Research Institute for Aerospace Medicine, Inha University, Incheon 22212, Republic of Korea; ^d Safety Measurement Institute, Korea Research Institute of Standards and Science; ^e Department of Medical Physics, Korea University of Science and Technology, Daejeon 34113 Republic of Korea.

ABSTRACT

Observing micro-vessels in conjunctiva could be used not only for diagnosing conjunctival diseases including conjunctivitis and pterygium but also as biomarkers for circulatory diseases. Many research teams have developed compact imaging and auto-analysis systems to simplify the conventional slit lamp as well as enhance the analysis process. The imaging system, previously developed by our research team, corrects eye motion in imaging windows through image registration and template matching. The developed system quantifies blood flow velocity using the sequence of motion-corrected images. This study compares estimated flow velocity and the fluid's actual velocity using the experimental phantom comprising transparent hoses and fluid including beads corresponding to red blood cells. The flow velocities are calculated using the Hagen-Poiseuille equation, and the flow rates generated by the syringe pump. The pump applies three kinds of flow rates to generate flow velocity variations and the estimated velocities are linearly proportional to these variations. In addition, the phantom has a random motion to mimic the fixational eye movements within the range of the healthy subject's angular eye motion. Through these experiments, we verified the previously developed flow velocity measurement system having percent errors under 3% by comparing estimated flow velocities with actual flow velocities. The system's accuracy, especially under conditions without artificial motion, is over 98.5%. These experiments can provide the supporting background for feasibility and accuracy in a further clinical study in conjunctival microcirculation.

Keywords: Conjunctival microvasculature, blood flow velocity, blood flow quantification, motion correction

1. INTRODUCTION

Conjunctiva is one of the most accessible tissues in humans to monitor the microvasculature due to the externally exposed and transparent mucous membrane covering on white sclera background. Monitoring conjunctival microvasculature (CM) is not confined as a method only to conjunctival disease including conjunctivitis, pterygium, and dry eye syndrome but could be used as biomarkers to check circulatory diseases [1-8]. Previous studies have demonstrated relations between blood flow velocities (BFV) in CM with various illnesses including diabetes, diabetic retinopathy, acute myocardial infarction, and brain diseases. Khansari, M.M., et al. reported BFV reduction is observed in subjects with proliferative diabetic retinopathy [5]. Valeshabad, A.K., et al. investigated unilateral ischemic stroke patients have slower BFV at the CM than in normal control groups [7]. Brennan, P.F., et al. demonstrated significantly reduced BFV for people with a recent acute myocardial infarction history [8]. We built a CM imaging system in the previous work for the feasibility studies of CM as a factor for monitoring patients' states [9]. This system acquired image frames using a high-magnification zoom lens and the complementary metal oxide semiconductor (CMOS) camera to capture red blood cells (RBC) with an average diameter of 7.5 μm [10]. The post-processing of the acquired images provided spatial quantification through convolutional neural network (CNN) based vessel segmentation and estimating morphological features of the vascular system. Flow velocities (FV) quantification tracked RBC movements based on the spatially processed image sequences. The spatial-temporal analysis (STA) as known as space time analysis, widely applied to quantify blood flows, provides spatial and temporal changes in RBC positions of fluid in the capillary [5-7, 11-13]. Although many studies including our

system have analyzed blood flows using the STA, demonstration of the phantom study as well as the reliability between results of estimation and actual flows are rare.

This study is the validating process of the previously built system by comparing actual and measured FV in vessel-mimicked imaging targets. The target is transparent tubes with fluid as blood substitutes composed of distilled water, microspheres, and red dye. Actual FV in the passages are calculated considering the flow rates of the syringe pump. In addition, the targets have artificial movements like clinical imaging conditions. Patients gazing at fixation targets have fixational eye movements such as drift, tremor, and microsaccade [14]. Irregularly controlled X-Y axis manual stages produce artificial motions substituting fixational eye movements on the the imaging targets.

2. METHODS AND MATERIALS

2.1 Imaging system

2.1.1 Imaging hardware and software

The custom-built optical imaging system in the previous study consisted of a CMOS camera, a magnification zoom lens, and a green LED (central wavelength 525nm) with a diffuser [9]. In addition, an aiming laser (CPS635R, Thorlabs Inc., Newton, NJ, USA) is recently added to adjust the quick position of the imaging system. The aiming laser points a rough position on the eyelid while the subjects close their eyes. The previous work described detailed information and part number of the components [9]. These components are fixed with optomechanics and adjusted by manual translation stages, as shown in Fig. 1(a).

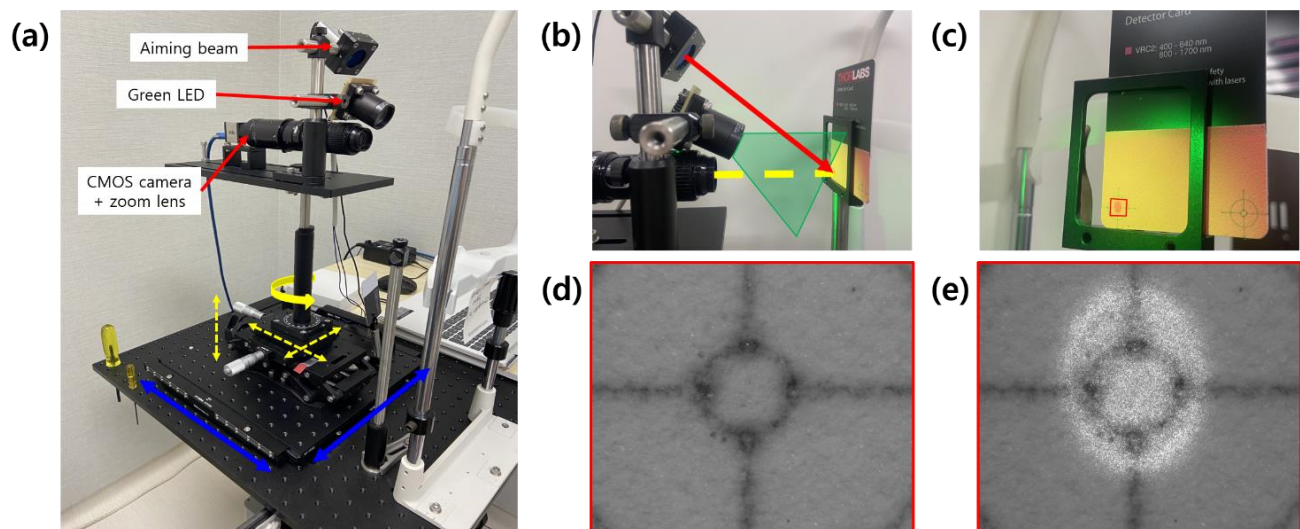


Figure 1. The custom-built optical imaging system. (a) an image of the whole hardware (yellow arrow means fine position adjustment and blue arrow means rough adjustment by manual stage), (b) the beam path of each part (imaging, green LED, and aiming beam), (c) imaging target and green LED illumination, (d) acquired image without aiming beam, (e) scattered beam by aiming laser in the acquired image. (d) and (e) are the image of the red box in (c).

The software, eye cockpit, served by the camera producer, IDS Inc., provides an acquisition function for 30 fps video data during fluid running in the passages. These data acquisition got frames with 2456 x 2054 pixels corresponding to the specific fields of view depending on the magnification value. Each pixel size is calculated with line pairs in an image of a USAF resolution target.

2.1.1 Flow velocities (FV) estimation

After the data acquisition, the FV are estimated through the five-step process in serial order of image registration, deep learning-based vessel segmentation, morphological feature extraction, template matching for motion correction, and spatial-temporal analysis.

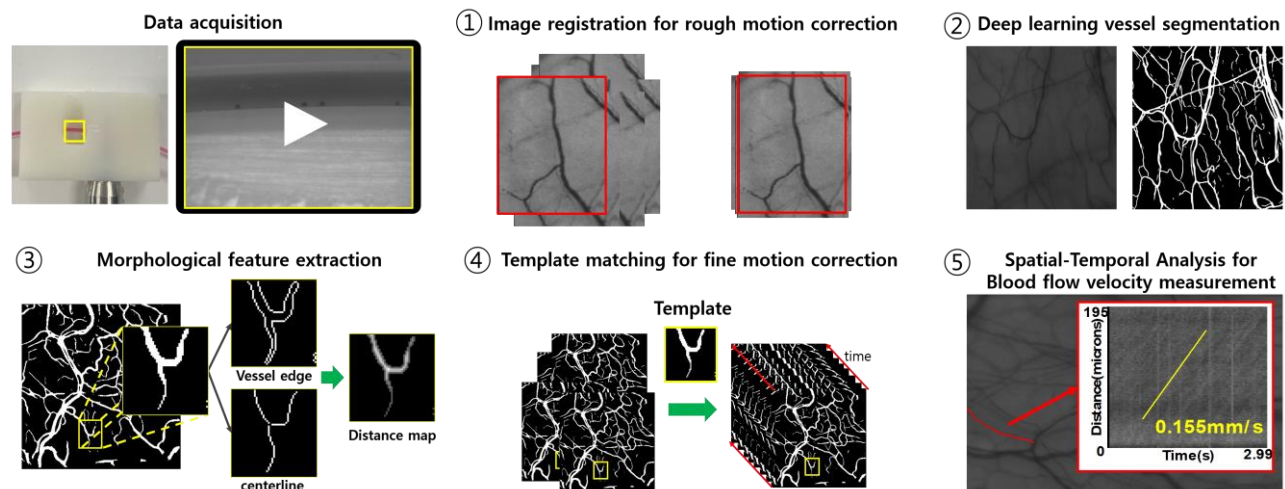


Figure 2. The progress of the FV estimation. This figure is reproduced of the previous paper [9].

Image registration is accomplished using the ImageJ plugin, MOCO (MOtion COrrector) which can align displaced frames to the template frame [15]. After the user designated a template frame, the MOCO automatically aligned the rest of the consecutive frames. This algorithm corrects image translation by maximizing the overlapping region between two images, thereby eliminating significant displacement of the images caused by motions. Vessel segmentation used a deep learning model trained by supervised learning of 240,000 patches from augmentation of the HRF dataset. The vessel segmentation process provides binary images of background and vasculature for easy morphological feature extraction. The skeletonized binary image for morphological feature extraction showed the vessel's edges which are used to get the centerline of the vessel by the pixel-wise thinning algorithm [16, 17]. Counting pixels of the centerline and between the centerline to the vessel's edge provides each vessel's morphological features, including the length and diameter. After getting these morphological features, a template image is assigned considering the following conditions. The designated template vessel must exist in all frames of the image sequence and must be distinctive from other vessels. The template matching algorithm shifts the rest of the images to template images for fine motion correction. Even though this template-matching process is fundamentally similar to the MOCO image registration, the algorithm and the reference image are distinctive. The cvMatchTemplate function of the OpenCV has been applied, and the segmented binary images that can be distinguished clearly have been used as a template and matching target images[18]. The corrected image sequences are obtained for spatial-temporal analysis (STA) through this motion correction process. In the image sequences, moving particles as a substitute for RBC in fluid mimicking blood make intensity differences in the passages, as shown in Fig 3(b, f). Gathering the intensity images of particles' path as a column during a specific time can produce a spatial-temporal analysis image, as shown in Fig 6(a). Like a linear graph, the diagonal pattern appears as a series of pixels for similar intensity values. The inclination of this pattern means the distance of the particle's flow per time as estimated as FV. The detailed principles and explanation of these five-step processes are described in the previous work ; for example, there are the formula of various image processing methods, data augmentation, learning and validation for deep learning model, and network architecture based on the convolutional neural network (CNN) [9].

2.2 Phantom experiments

We used transparent tubes as vessel substitutes and fluids with microspheres as RBC substitutes to mimic the vascular system. The microspheres are the target for tracking fluid flow and estimating FV. This phantom study is performed under two conditions with and without artificial movements on the phantom. The actual FV in the phantom is estimated and

calculated by the Hagen-Poiseuille equation using the flow rate of the syringe pump. The input of the fluid mixture in phantoms is connected to the syringe on the syringe pump, which makes specific and constant flow.

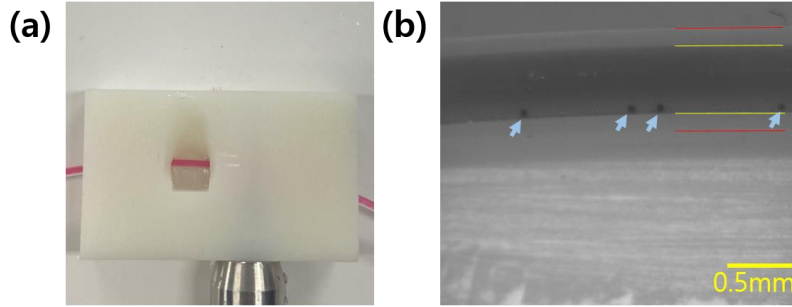


Figure 3. Fluidic phantoms used in these experiments. (a) tube phantom in 3D-printed housing, (b) acquired image of flow with dark red microspheres (diameters: 100 μm), Yellow line presented inner boundary and the red line gave outer boundary. Sky blue arrows present microspheres in (b).

2.2.1 Flow velocity (FV) calculation

In this experiment, a transparent tube made of Polytetrafluoroethylene (PTFE)(11-2559, Sciencix, Burnsville, MN, USA) is used to simulate the flow of blood through a blood vessel. The tube has an inner diameter of 770 μm and the fluid has dark red color microspheres with 100 μm (56969, Sigma Aldrich, St. Louis, MO, USA). The microspheres in the fluid are imaged, as shown in Figure 3(b). The syringe pump made a pressure with constant flow rates to flow the mixture of red watercolors, dark red microspheres, and distilled water. According to the Hagen-Poiseuille equation, the FV of fluid with viscosity differs depending on the distance from the axis of the tube, as presented in Fig. 4.

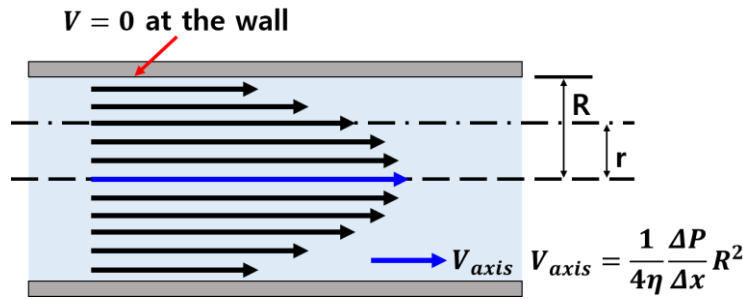


Figure 4. Flow velocities of fluid with viscosity depending on the distance from the tube center. R is the inner radius of the tube, r is the distance from the axis, η is the viscosity of the fluid, P is pressure applied for fluid, and x is the length of flow.

Calculating the FV of the region as far as "r" from the axis is crucial for the comparison with estimated velocities in the finding of actual FV. FV profiles away from the axis are calculated in equation (1).

$$V(r) = V_{axis} \left[1 - \frac{r^2}{R^2} \right] \quad \text{..... (1)}$$

The flow rate ω is presented as equation (2). Through equations (2, 3), we can know the V_{axis} because the flow rate is constantly maintained under experimental conditions. Applying the equation for the V_{axis} on the equation (2), the flow rate is substituted to equation (3).

$$\omega = \frac{\pi}{8\eta} \frac{\Delta P}{\Delta x} R^4 \quad \text{..... (2)}$$

$$\omega = V_{axis} \times \frac{1}{2} \pi R^2 \quad \text{..... (3)}$$

2.2.2 Artificial motion generation

It is hard to simulate perfectly fixational eye movements and clinical conditions, but generating displacement is possible using adjustable stages. According to the previous research for fixational eye movements, microsaccade, which is the most crucial motion for stable imaging, is moved in the range of -1.5 degrees to 1.5 degrees in healthy subjects[19]. Moreover, patients with ophthalmic diseases have microsaccades ranging between -2 degrees to 3 degrees[19]. These angular movements have to be transformed into a flat plane because the relatively displaced vessel patterns are imaged as a plane. The focal plane for imaging and actual vessel pattern displacements are presented on the specific flat plane, which is described in Fig 5. Although the human eyeball is originally not a perfect sphere but an elliptical sphere with variations, the transform equation uses an eyeball as a sphere with a diameter of 24.5 mm for "R"[20]. Based on this distortion, we transformed the angular range of the fixational eye movements to a millimeter unit (table 1). In this experiment, the artificial motion range is within the healthy subject's fixational eye movements because the displacement range of diseased subjects can dramatically reduce the corrected imaging window after the motion correction.

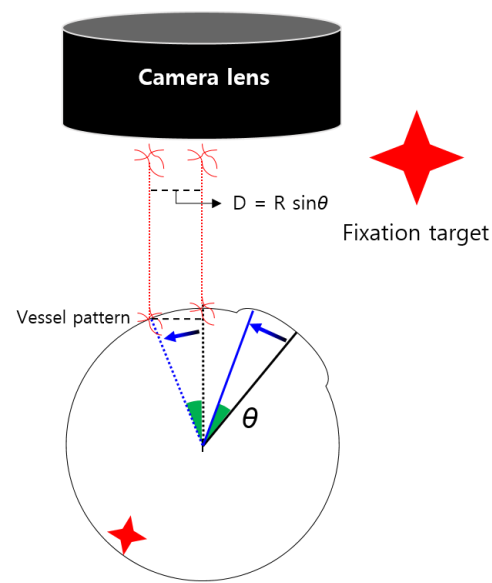


Figure 5. Schematic of transform for angular movement

Table 1. Ranges of fixational eye movement and transformed contents as millimeter unit

	Healthy subject	Strabismic amblyopia	Macular disease
Range of (micro)saccades (°)	-1.5 ~ 1.5 [19]	0 ~ 5 [21]	-2 ~ 3 [19]
Transformed range for camera sight (mm)	-0.641 ~ 0.641	0 ~ 2.125	-0.855 ~ 1.282

3. RESULTS

3.1 Flow velocities(FV) of phantoms without artificial motion

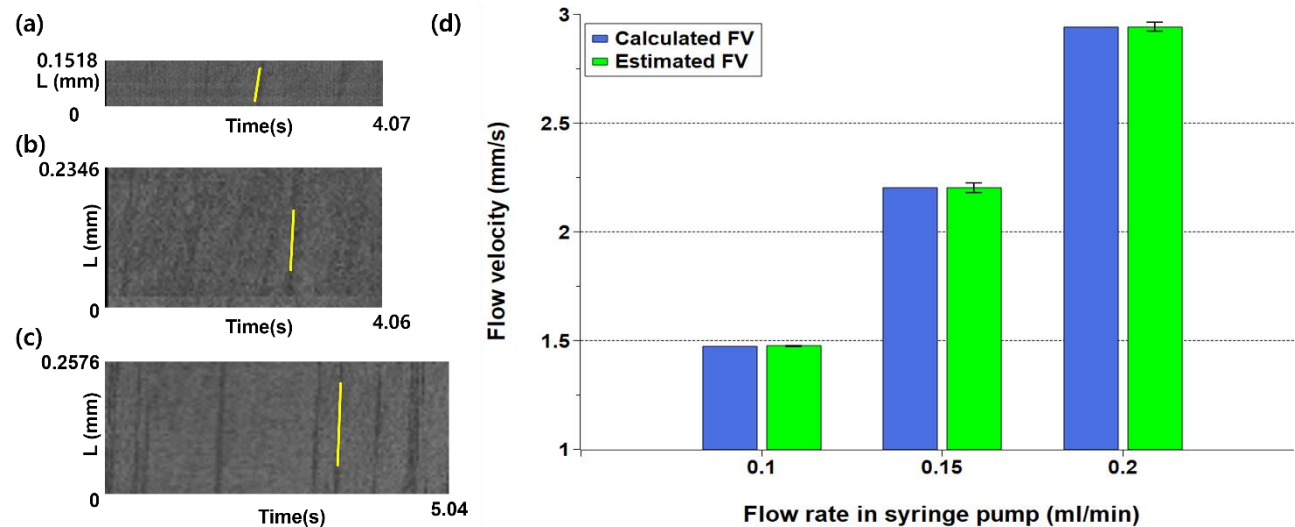


Figure 6. Flow velocities (FV) estimation of tube phantom. (a) STA image of the flow rates 0.1ml/min (b). The yellow line shows the focusing area for estimating FV, (c) STA image of the flow rates 0.15ml/min, (d) comparison between calculated and estimated FV. Error bars are standard deviations of estimated FV.

The flow rates ω , generated by the syringe pump, are 0.1ml/min, 0.15ml/min, and 0.2ml/min in the tube with a diameter of 770 μm . The FV of the region for microspheres' moving (343.2 μm away from the axis) is theoretically 1.47 mm/s (for 0.1ml/min), 2.20 mm/s (for 0.15ml/min), and 2.94 mm/s (for 0.2ml/min) which are calculated via equation (1). Fig. 6(a, b, c) shows STA data of one of the measurement data in each flow rate. The inclination of the yellow graph means flow velocity. Fig 6(d) presents the calculated FV and estimated FV. Average (AVG) of estimated FVs are get through 5 times experiments for each flow rate. Table 2. is detailed FV data that we get. The percent error of our estimation method is under 1.5% and the standard deviation(STD) is under 0.025 in this study.

Table 2. Flow velocities (FV) by each method and percentage errors between methods.

Flow rate (ml/min)	CFV (mm/s)	EFV (mm/s)		(CFV-EFV / CFV) %
0.1	1.4699	1.4715		0.11%
0.1	1.4699	1.4739		0.27%
0.1	1.4699	1.4732		0.22%
0.1	1.4699	1.4744		0.30%
0.1	1.4699	1.4752		0.36%
		AVG	1.4736	
		STD	0.0014	
0.15	2.2049	2.1775		1.25%
0.15	2.2049	2.1748		1.37%
0.15	2.2049	2.2152		0.47%
0.15	2.2049	2.2179		0.59%
0.15	2.2049	2.2172		0.55%
		AVG	2.2005	
		STD	0.0223	
0.2	2.9399	2.9668		0.91%
0.2	2.9399	2.9433		0.11%
0.2	2.9399	2.9233		0.57%
0.2	2.9399	2.9194		0.70%
0.2	2.9399	2.9502		0.35%
		AVG	2.9406	
		STD	0.0196	

3.2 Flow velocities(FV) of phantoms with artificial motion

The fixational eye movement has disturbed conjunctival imaging in clinical conditions. The normal STA without disturbance by motion can show the series of the pixel intensity of estimating region depending on the frame sequence like Fig 7(a). If the imaging system does not correct motion artifacts, the STA image for FV estimation is destroyed as shown in Fig 7(b). The destroyed STA image is not able to provide proper FV data.

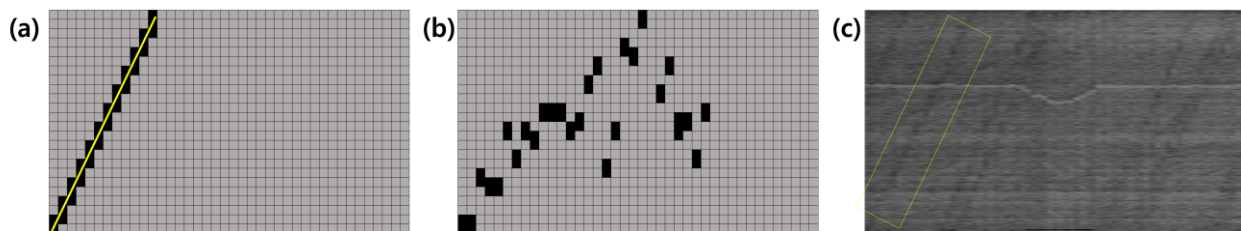


Figure 7. Explanation of STA image. (a) graphic for describing a normal STA image without motion artifacts, (b) graphic for describing a distorted STA image by motion artifact, (c) acquired STA image in these experiments.

Two-step algorithm for motion corrections, MOCO, and cvMatchTemplate, corrected motion artifacts through pixel shifting of each frame. After processing the motion-corrected image sequence, the imaging system provides STA images with line patterns meaning flow velocity, as shown in Fig 7(c). To validate the quantification function of flow velocities in the system operating condition with motion artifacts, the fluid run at the tube at a 0.2ml/min flow rate, and housing is manually displaced by X-Y stages. If the motion correction algorithm is valid, the estimated flow velocity has to meet the

near 2.9399 mm/s. Fig 8. provides frame movement value for motion correction in each algorithm, an estimated region for FV, and comparisons calculated FV with estimated FV after motion correction.

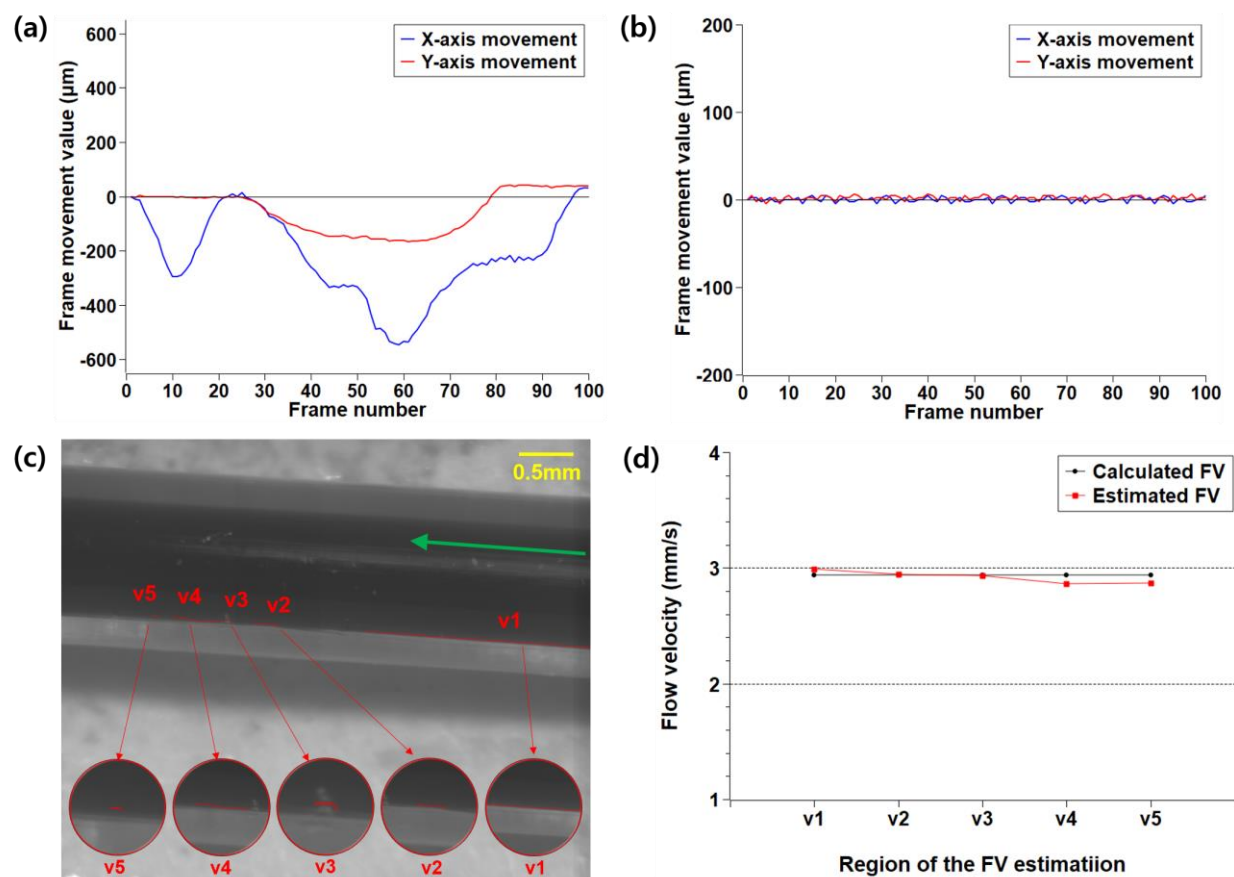


Figure 8. Flow velocities (FV) estimation of tube phantom with artificial motion. (a) frame movements by MOCO algorithm, (b) frame movements by cvMatchTemplate algorithm after MOCO, (c) image of presenting the region for FV estimation as red pixels. The green arrow presents the flow direction, (c) FV of each estimating method.

Table 3. Flow velocities (FV) of tube phantom with artificial motion and percentage errors between methods. C is CFV, E is EFV, and M is MFV.

Region of the FV estimation	CFV(mm/s)	EFV(mm/)	(CFV-EFV / CFV) %
V1	2.9399	2.9914	1.75
V2	2.9399	2.9483	0.29
V3	2.9399	2.9326	0.25
V4	2.9399	2.8601	2.71
V5	2.9399	2.8713	2.33
		AVG	2.9207
		STD	0.0548

Estimated FVs are the FV at each region for estimation. In this case, artificial motion is randomly generated by manual control, repeat experiments could be not performed. Table 3. is detailed FV data that we get. The percent error of our estimation method is under 3.0% and the standard deviation(STD) is under 0.055 in this study.

4. CONCLUSION

In this study, we investigated the validation of a customized conjunctival blood flow velocity measurement system through the phantom study. The phantom is transparent tubes, as well as the fluidic mixture, consisting of red watercolors, distilled water, and dark red color microspheres corresponding to red blood cells. The syringe pump produced constant pressure to make specific flow velocities during the image sequence acquisition. The comparison between the estimated and calculated flow velocities provides the measurement system's percent errors. The accuracy of flow velocity measurement is over 98.5% at the three flow rates in the vascular systems of each experiment. In addition, this accuracy is validated under the being disturbed by motion artifacts via the phantoms with artificially generated parallel movement. The accuracy is over 97% in the condition of the target having parallel motions. These experiments can provide the supporting background for feasibility and accuracy in a further clinical study in conjunctival microcirculation.

5. ACKNOWLEDGEMENTS

This research was supported by the National Research Foundation of Korea (2018R1A6A1A03025523) and by the Korea Medical Device Development Fund grant funded by the Korea government (the Ministry of Science and ICT, the Ministry of Trade, Industry and Energy, the Ministry of Health & Welfare, the Ministry of Food and Drug Safety) (Project Number: KMDF_PR_20200901_0026)

REFERENCES

- [1] W. Chen, H. I. M. Batawi, J. R. Alava *et al.*, "Bulbar conjunctival microvascular responses in dry eye," *The ocular surface*, 15(2), 193-201 (2017).
- [2] J. Wang, H. Jiang, A. Tao *et al.*, "Limbal capillary perfusion and blood flow velocity as a potential biomarker for evaluating dry eye," *Investigative Ophthalmology & Visual Science*, 54(15), 4335-4335 (2013).
- [3] W. Chen, Y. Deng, H. Jiang *et al.*, "Microvascular abnormalities in dry eye patients," *Microvascular research*, 118, 155-161 (2018).
- [4] V. C. Karanam, L. Tamariz, H. Batawi *et al.*, "Functional slit lamp biomicroscopy metrics correlate with cardiovascular risk," *The ocular surface*, 17(1), 64-69 (2019).
- [5] M. M. Khansari, J. Wanek, M. Tan *et al.*, "Assessment of conjunctival microvascular hemodynamics in stages of diabetic microvasculopathy," *Scientific reports*, 7(1), 1-9 (2017).
- [6] P. F. Brennan, A. J. McNeil, M. Jing *et al.*, "Quantitative assessment of the conjunctival microcirculation using a smartphone and slit-lamp biomicroscope," *Microvascular research*, 126, 103907 (2019).
- [7] A. K. Valeshabad, J. Wanek, F. Mukarram *et al.*, "Feasibility of assessment of conjunctival microvascular hemodynamics in unilateral ischemic stroke," *Microvascular research*, 100, 4-8 (2015).
- [8] P. F. Brennan, A. J. McNeil, M. Jing *et al.*, "Assessment of the conjunctival microcirculation for patients presenting with acute myocardial infarction compared to healthy controls," *Scientific Reports*, 11(1), 7660 (2021).
- [9] H.-C. Jo, H. Jeong, J. Lee *et al.*, "Quantification of Blood Flow Velocity in the Human Conjunctival Microvessels Using Deep Learning-Based Stabilization Algorithm," *Sensors*, 21(9), 3224 (2021).
- [10] E. L. Persons, "Studies on red blood cell diameter: III. The relative diameter of immature (reticulocytes) and adult red blood cells in health and anemia, especially in pernicious anemia," *The Journal of clinical investigation*, 7(4), 615-629 (1929).
- [11] S. P. Chong, C. W. Merkle, C. Leahy *et al.*, "Quantitative microvascular hemoglobin mapping using visible light spectroscopic Optical Coherence Tomography," *Biomedical optics express*, 6(4), 1429-1450 (2015).
- [12] M. Shahidi, J. Wanek, B. Gaynes *et al.*, "Quantitative assessment of conjunctival microvascular circulation of the human eye," *Microvascular research*, 79(2), 109-113 (2010).
- [13] M. M. Khansari, J. Wanek, A. E. Felder *et al.*, "Automated assessment of hemodynamics in the conjunctival microvasculature network," *IEEE transactions on medical imaging*, 35(2), 605-611 (2015).
- [14] J. Otero-Millan, S. L. Macknik, and S. Martinez-Conde, "Fixational eye movements and binocular vision," *Frontiers in integrative neuroscience*, 8, 52 (2014).

- [15] A. Dubbs, J. Guevara, and R. Yuste, "moco: Fast motion correction for calcium imaging," *Frontiers in neuroinformatics*, 10, 6 (2016).
- [16] T.-C. Lee, R. L. Kashyap, and C.-N. Chu, "Building skeleton models via 3-D medial surface axis thinning algorithms," *CVGIP: Graphical Models and Image Processing*, 56(6), 462-478 (1994).
- [17] J. W. Klingler, C. L. Vaughan, T. Fraker *et al.*, "Segmentation of echocardiographic images using mathematical morphology," *IEEE Transactions on Biomedical Engineering*, 35(11), 925-934 (1988).
- [18] G. Bradski, and A. Kaehler, [Learning OpenCV: Computer vision with the OpenCV library] " O'Reilly Media, Inc.", (2008).
- [19] G. Kumar, and S. T. Chung, "Characteristics of fixational eye movements in people with macular disease," *Investigative ophthalmology & visual science*, 55(8), 5125-5133 (2014).
- [20] I. Bekerman, P. Gottlieb, and M. Vaiman, "Variations in eyeball diameters of the healthy adults," *Journal of ophthalmology*, 2014, (2014).
- [21] E. G. González, A. M. Wong, E. Niechwiej-Szwedo *et al.*, "Eye position stability in amblyopia and in normal binocular vision," *Investigative ophthalmology & visual science*, 53(9), 5386-5394 (2012).